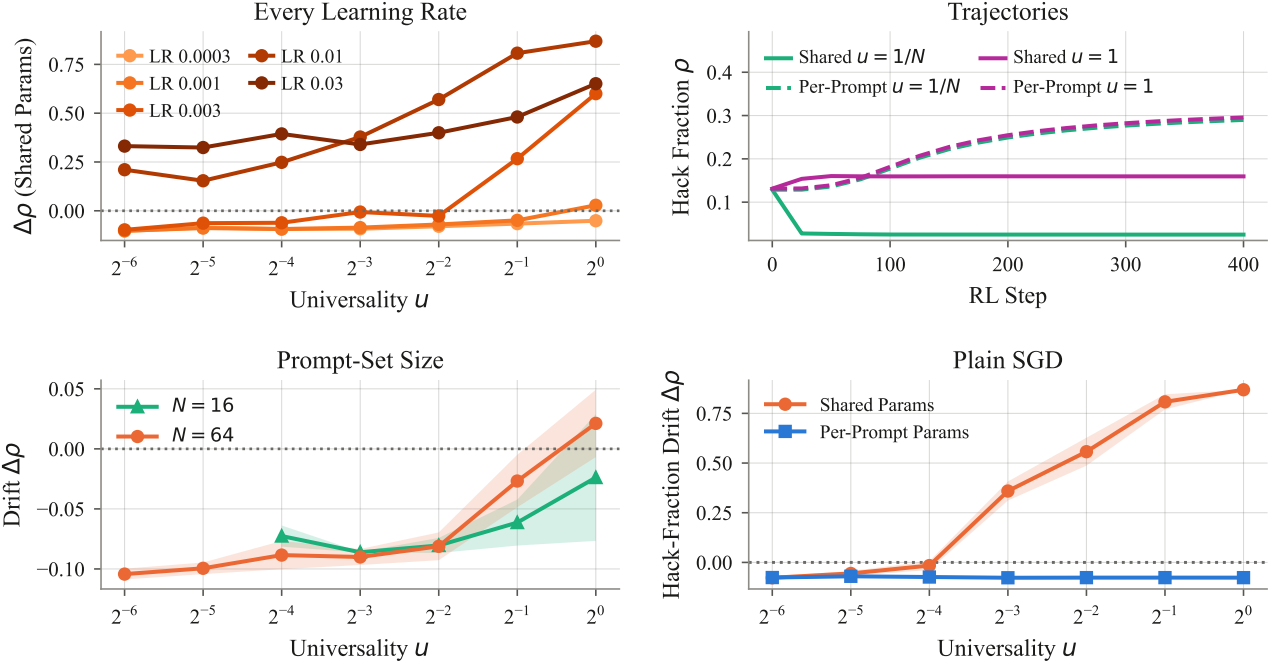
Every Learning Rate
LR 0.0003
LR 0.001
LR 0.003
LR 0.01
LR 0.03
Δρ (Shared Params)
0.75
0.50
0.25
0.00
2^−6
2^−5
2^−4
2^−3
2^−2
2^−1
2^0
Universality u
Trajectories
Shared u = 1/N
Per-Prompt u = 1/N
Shared u = 1
Per-Prompt u = 1
Hack Fraction ρ
0.4
0.3
0.2
0.1
0
100
200
300
400
RL Step
Prompt-Set Size
N = 16
N = 64
Drift Δρ
0.05
0.00
−0.05
−0.10
Universality u
Plain SGD
Shared Params
Per-Prompt Params
Hack-Fraction Drift Δρ
0.75
0.50
0.25
0.00
Universality u